Cancer

Diagnosis

Cancer screening

Diagnosing cancer at its earliest stages often provides the best chance for a cure. With this in mind, talk with your doctor about what types of cancer screening may be appropriate for you.

For a few cancers, studies show that screening tests can save lives by diagnosing cancer early. For other cancers, screening tests are recommended only for people with increased risk.

A variety of medical organizations and patient-advocacy groups have recommendations and guidelines for cancer screening. Review the various guidelines with your doctor and together you can determine what's best for you based on your own risk factors for cancer.

Cancer diagnosis

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Your doctor may use one or more approaches to diagnose cancer:

- **Physical exam.** Your doctor may feel areas of your body for lumps that may indicate cancer. During a physical exam, your doctor may look for abnormalities, such as changes in skin color or enlargement of an organ, that may indicate the presence of cancer.
- **Laboratory tests.** Laboratory tests, such as urine and blood tests, may help your doctor identify abnormalities that can be caused by cancer. For instance, in people with leukemia, a common blood test called complete blood count may reveal an unusual number or type of white blood cells.
- **Imaging tests.** Imaging tests allow your doctor to examine your bones and internal organs in a noninvasive way. Imaging tests used in diagnosing cancer may include a computerized tomography (CT) scan, bone scan, magnetic resonance imaging (MRI), positron emission tomography (PET) scan, ultrasound and X-ray, among others.
- **Biopsy.** During a biopsy, your doctor collects a sample of cells for testing in the laboratory. There are several ways of collecting a sample. Which biopsy procedure is right for you depends on your type of cancer and its location. In most situations, a biopsy is the only way to definitively diagnose cancer.

  In the laboratory, doctors look at cell samples under the microscope. Normal cells look uniform, with similar sizes and orderly organization. Cancer cells look less orderly, with varying sizes and without apparent organization.

Cancer stages

Once cancer is diagnosed, your doctor will work to determine the extent (stage) of your cancer. Your doctor uses your cancer's stage to determine your treatment options and your chances for a cure.

Staging tests and procedures may include imaging tests, such as bone scans or X-rays, to see if cancer has spread to other parts of the body.

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Cancer stages are indicated by the numbers 0 through 4, which are often written as Roman numerals 0 through IV. Higher numbers indicate a more-advanced cancer. For some types of cancer, cancer stage is indicated using letters or words.

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Treatment

Many cancer treatments are available. Your treatment options will depend on several factors, such as the type and stage of your cancer, your general health, and your preferences. Together you and your doctor can weigh the benefits and risks of each cancer treatment to determine which is best for you.

Cancer Clinical Trials Offer Many Benefits

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Goals of cancer treatment

Cancer treatments have different objectives, such as:

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- **Cure.** The goal of treatment is to achieve a cure for your cancer, allowing you to live a normal life span. This may or may not be possible, depending on your specific situation.
- **Primary treatment.** The goal of a primary treatment is to completely remove the cancer from your body or kill the cancer cells.

  Any cancer treatment can be used as a primary treatment, but the most common primary cancer treatment for the most common cancers is surgery. If your cancer is particularly sensitive to radiation therapy or chemotherapy, you may receive one of those therapies as your primary treatment.
- **Adjuvant treatment.** The goal of adjuvant therapy is to kill any cancer cells that may remain after primary treatment in order to reduce the chance that the cancer will recur.

  Any cancer treatment can be used as an adjuvant therapy. Common adjuvant therapies include chemotherapy, radiation therapy and hormone therapy.
- **Palliative treatment.** Palliative treatments may help relieve side effects of treatment or signs and symptoms caused by cancer itself. Surgery, radiation, chemotherapy and hormone therapy can all be used to relieve symptoms and control the spread of cancer when a cure isn't possible. Medications may relieve symptoms such as pain and shortness of breath.

  Palliative treatment can be used at the same time as other treatments intended to cure your cancer.

Cancer treatments

Doctors have many tools when it comes to treating cancer. Cancer treatment options include:

- **Surgery.** The goal of surgery is to remove the cancer or as much of the cancer as possible.

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- **Chemotherapy.** Chemotherapy uses drugs to kill cancer cells.
- **Radiation therapy.** Radiation therapy uses high-powered energy beams, such as X-rays and protons, to kill cancer cells. Radiation treatment can come from a machine outside your body (external beam radiation), or it can be placed inside your body (brachytherapy).
- **Bone marrow transplant.** Bone marrow transplant is also known as a stem cell transplant. Your bone marrow is the material inside your bones that makes blood cells. A bone marrow transplant can use your own cells or cells from a donor.

  A bone marrow transplant allows your doctor to use higher doses of chemotherapy to treat your cancer. It may also be used to replace diseased bone marrow.
- **Immunotherapy.** Immunotherapy, also known as biological therapy, uses your body's immune system to fight cancer. Cancer can survive unchecked in your body because your immune system doesn't recognize it as an intruder. Immunotherapy can help your immune system "see" the cancer and attack it.
- **Hormone therapy.** Some types of cancer are fueled by your body's hormones. Examples include breast cancer and prostate cancer. Removing those hormones from the body or blocking their effects may cause the cancer cells to stop growing.
- **Targeted drug therapy.** Targeted drug treatment focuses on specific abnormalities within cancer cells that allow them to survive.
- **Clinical trials.** Clinical trials are studies to investigate new ways of treating cancer. Thousands of cancer clinical trials are underway.

Other treatments may be available to you, depending on your type of cancer.

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Clinical trials

Explore Mayo Clinic studies testing new treatments, interventions and tests as a means to prevent, detect, treat or manage this condition.

Alternative medicine

No alternative cancer treatments have been proved to cure cancer. But alternative medicine options may help you cope with side effects of cancer and cancer treatment, such as fatigue, nausea and pain.

Talk with your doctor about which alternative medicine options may offer some benefit. Your doctor can also discuss whether these therapies are safe for you or whether they may interfere with your cancer treatment.

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Some alternative medicine options found to be helpful for people with cancer include:

- Acupuncture
- Hypnosis
- Massage
- Meditation
- Relaxation techniques
- Yoga

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High-dose vitamin C: Can it kill cancer cells?

Coping and support

A cancer diagnosis can change your life forever. Each person finds his or her own way of coping with the emotional and physical changes cancer brings. But when you're first diagnosed with cancer, sometimes it's difficult to know what to do next.

Here are some ideas to help you cope:

- **Learn enough about cancer to make decisions about your care.** Ask your doctor about your cancer, including your treatment options and, if you like, your prognosis. As you learn more about cancer, you may become more confident in making treatment decisions.

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- **Keep friends and family close.** Keeping your close relationships strong will help you deal with your cancer. Friends and family can provide the practical support you'll need, such as helping take care of your house if you're in the hospital. And they can serve as emotional support when you feel overwhelmed by cancer.
- **Find someone to talk with.** Find a good listener who is willing to listen to you talk about your hopes and fears. This may be a friend or family member. The concern and understanding of a counselor, medical social worker, clergy member or cancer support group also may be helpful.

  Ask your doctor about support groups in your area. Other sources of information include the National Cancer Institute and the American Cancer Society.

Preparing for your appointment

Start by making an appointment with your doctor if you have any signs or symptoms that worry you. If your doctor determines that you have cancer, you'll likely be referred to one or more specialists, such as:

- Doctors who treat cancer (oncologists)
- Doctors who treat cancer with radiation (radiation oncologists)
- Doctors who treat diseases of the blood and blood-forming tissues (hematologists)
- Surgeons

Because appointments can be brief, and because there's often a lot information to discuss, it's a good idea to be prepared. Here's some information to help you get ready, and know what to expect from your doctor.

What you can do

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- **Be aware of any pre-appointment restrictions.** At the time you make the appointment, be sure to ask if there's anything you need to do in advance, such as restrict your diet.
- **Write down any symptoms you're experiencing,** including any that may seem unrelated to the reason for which you scheduled the appointment.
- **Write down key personal information,** including any major stresses or recent life changes.
- **Write down your family's history of cancer.** If other members of your family have been diagnosed with cancer, make a note of the types of cancer, how each person is related to you and how old each person was when diagnosed.
- **Make a list of all medications,** vitamins or supplements that you're taking.
- **Consider taking a family member or friend along.** Sometimes it can be difficult to remember all the information provided during an appointment. Someone who accompanies you may remember something that you missed or forgot.
- **Write down questions to ask** your doctor.

Your time with your doctor is limited, so preparing a list of questions can help you make the most of your time together. List your questions from most important to least important in case time runs out. For cancer, some basic questions to ask your doctor include:

- What type of cancer do I have?
- What stage is my cancer?
- Will I need additional tests?
- What are my treatment options?
- Can treatments cure my cancer?
- If my cancer can't be cured, what can I expect from treatment?

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- What are the potential side effects of each treatment?
- Is there one treatment you feel is best for me?
- How soon do I need to begin treatment?
- How will treatment affect my daily life?
- Can I continue working during treatment?
- Are there any clinical trials or experimental treatments available to me?
- I have these other health conditions. How can I manage them during my cancer treatment?
- Are there any restrictions that I need to follow?
- Should I see a specialist? What will that cost, and will my insurance cover it?
- Is there a generic alternative to the medicine you're prescribing?
- Are there brochures or other printed material that I can take with me? What websites do you recommend?
- What will determine whether I should plan for follow-up visits?

In addition to the questions that you've prepared to ask your doctor, don't hesitate to ask other questions that occur to you.

What to expect from your doctor

Your doctor is likely to ask you a number of questions. Being ready to answer them may allow time later to cover other points you want to address. Your doctor may ask:

- When did you first begin experiencing symptoms?
- Have your symptoms been continuous or occasional?
- How severe are your symptoms?
- What, if anything, seems to improve your symptoms?

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- What, if anything, appears to worsen your symptoms?
- Does anyone in your family have cancer?
- Have you ever had cancer before? If so, what kind and how was it treated?
- Have you ever been exposed to chemicals at home or at work?
- Do you smoke or use tobacco?
- Have you ever been diagnosed with a hepatitis infection or a human papillomavirus infection?

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Living with cancer?

Connect with others like you for support and answers to your questions in the Cancer support group on Mayo Clinic Connect, a patient community.

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